

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457072.84092 +/- 0.00088 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.028
Transit depth (Rp/Rs)^2	3.89 +/- 1.11 [%]
Orbital Inclination (inc)	82.01 +/- 1.96
Airmass coefficient 1 (a1)	0.672 +/- 0.023
Airmass coefficient 2 (a2)	0.287 +/- 0.09
Scatter in the residuals of the lightcurve fit is	3.39 %
Best Comparison Star	#3 - [67, 57]
Optimal Aperture	3.45
Optimal Annulus	5.35
Transit Duration (day)	0.0522 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.197 ± 0.028 vs 0.1594 (deviation 1.18σ)
Duration fit vs published	0.0522 d vs 0.0512 d (deviation 0.12σ)
Mid-time O–C	-2.3 min
Depth SNR	6.2
Residual scatter	3.39 %

Light curve

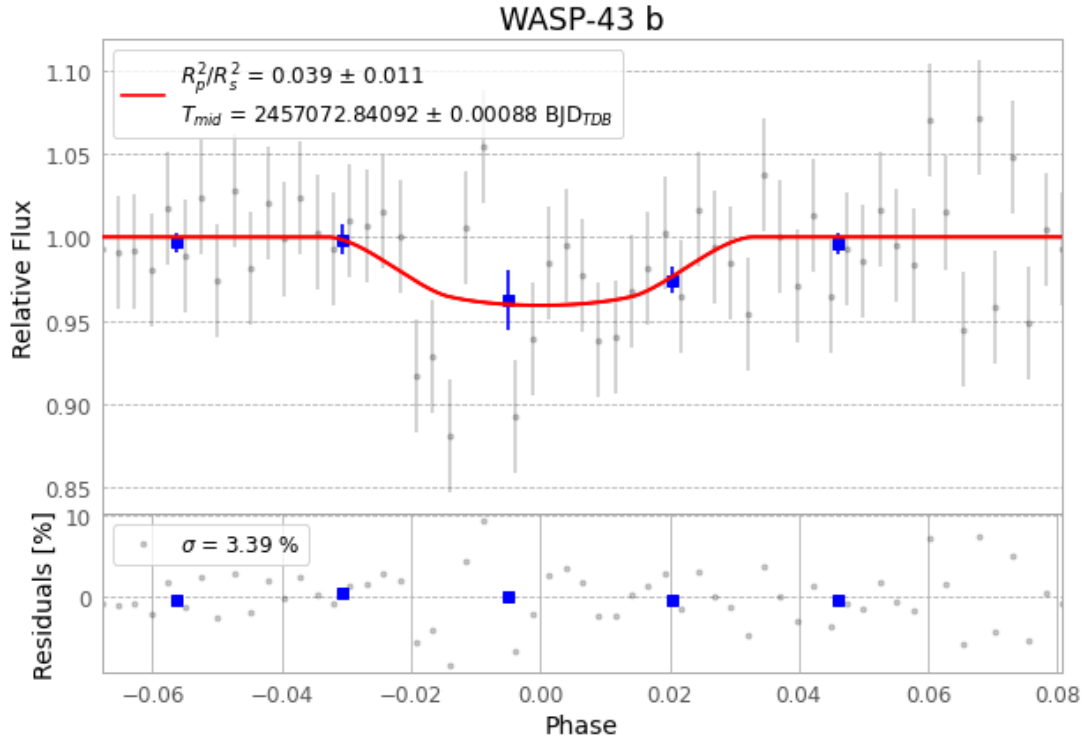


Figure 1 — FinalLightCurve_WASP-43 b_2015-02-18.png (SHA-256 420970f254564bae...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [153, 224], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3798fed972047540...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43150219064212.FITS → WASP-43150219095112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-150219.log (5ba14d235f99...), FinalLightCurve_WASP-43 b_2015-02-18.png (420970f25456...), FinalLightCurve_WASP-43 b_2015-02-18.csv (53ba4a74e56b...)

Compute resources

Wall time 554.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 00:02Z.